

Kinematics

1 important concepts

see *Newtons Laws*

always label the axis that you are using to referenced

always justify any equations and specify any assumptions, i.e. that acceleration is constant

CONSTANT SPEED DOESN'T MEAN NO ACCELERATION, AS CIRCULARLY THERE IS CENTRIPETAL ACCELERATION FOR CONSTANT TANGENTIAL SPEED.

[W] refers to the units of Wb

some standard units

$$\vec{v} = LT^{-1}$$

$$E = mc^2$$

$$= [m][LT^{-1}]^2$$

We typically describe position, velocity, and acceleration with a set of basis vectors.

We can define the basis vectors as any pair of axis that are not on the same span, allowing for some combination of that vector to find any coordinate. These basis vectors are typically expressed

$$\hat{x} \quad \hat{y} \quad \hat{z}$$

we should think of velocity as an expanded product rule. For cartesian basis vectors that don't change over time,

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}x(t)\hat{x} + \frac{d}{dt}y(t)\hat{y} \quad (1)$$

which can be expanded by product rule for derivatives along each basis vector

tricky notes in notation: #notation

any vector quantity requires the vector sign $||\vec{x}|| = |\vec{x}| = x$ which is the magnitude.

a_0 represents a constant

2 Parabolic motion

we can say that

$$\vec{r} = c_1 t \hat{x} + (c_2 t - c_3 t^2) \hat{y}$$

we can see that $x(t)$ is given by a constant c_1
and $y(t)$ is given by a function

$$(c_2 - 2c_3 t)$$

we can find the acceleration to be a constant

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = -2c_3 \hat{y} = g = \frac{9.8m}{s^2}$$

If instead,, we wanted to go from acceleration to position, then it is a more complicated story.
for instance, there are some interesting differential equations that appear throughout physics.

$$\vec{a} = -c_1 \vec{v} + \vec{c}_2 \quad \vec{a} = -c_2 v_1 \vec{r} + c_2 \vec{r}$$

If we assume that we have uniform acceleration, then we can integrate to get velocity, and integrate again to get position over time.

$$\begin{aligned} dx &= V_x(t) \\ &= v_0 + a_0 t \end{aligned}$$

which yields $\int_0^t (v_0 + a_0 t) dt$

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t, \vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a}t^2$$

3 circular motion

We also have a rotational equivalent of speed and acceleration. Given θ as a position, ω as an angular velocity, and α as angular acceleration. $\theta = \omega t$ describes uniform circular motion. in a circle there are 2π radians that can be traveled across. We have an exact equivalent to linear motion, using x position as angle. The magnitude of a uniform circular motion is R , and the acceleration is always radially inwards towards the center of the circle as $-\omega^2 \times R$

For Uniform circular motion,

$$v = \omega r \tag{2}$$

$$a_c = \omega^2 r = \frac{v^2}{r} \tag{3}$$

$$\theta = \omega t \tag{4}$$

see next: *Newtons Laws*
reference: *Formulas*

#kinematics
#circular_motion
#uniform_motion
#physics